

Exercise 2: E-commerce Platform Search Function

```
using System;
```

```
using System.Collections.Generic;
```

```
class Product
```

```
{
```

```
    public int ProductID;
```

```
    public string ProductName;
```

```
    public string Category;
```

```
    public Product(int productID, string productName, string category)
```

```
    {
```

```
        ProductID = productID;
```

```
        ProductName = productName;
```

```
        Category = category;
```

```
    }
```

```
}
```

```
class Program
```

```
{
```

```
    static int LinearSearch(List<Product> products, int key)
```

```
    {
```

```
        for (int i = 0; i < products.Count; i++)
```

```
        {
```

```
            if (products[i].ProductID == key)
```

```
                return i;
```

```
        }
```

```
        return -1;
```

```
    }
```

```

static int BinerySearch(List<Product> products , int len ,int key)
{
    int low = 0;
    int high = len - 1;
    int mid = (low + high) / 2;

    while(low<=high)
    {
        mid = (low + high) / 2;
        if (products[mid].ProductID == key)
            return mid;
        else if (key>= products[mid].ProductID)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}

```

```

static void Main(string[] args)
{
    List<Product> products = new List<Product>()
    {
        new Product(101, "iPhone", "Mobile"),
        new Product(102, "HP", "Laptop"),
        new Product(103, "BoAt", "Earphones"),
        new Product(104, "Titan", "Watch"),
        new Product(105, "iPad", "Tab")
    };

    int key = 103;
}

```

```
int index = LinearSearch(products, key);

if (index != -1)
{
    Console.Write("Product Found:");
    Console.WriteLine(products[index].ProductName);
}
else
{
    Console.WriteLine("Product Not Found");
}

int len = products.Count;
index = BinarySearch(products, len, 103);
if (index != -1)
{
    Console.Write("Product Found:");
    Console.WriteLine(products[index].ProductName);
}
else
{
    Console.WriteLine("Product Not Found");
}
}
```

```
Microsoft Visual Studio Debu  x + v
Product Found:BoAt
Product Found:BoAt
C:\Users\HP\source\repos\MyProgram\MyProgram\bin\Debug\net10.0\MyProgram.exe (process 1136) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```